HYDROGELS

Soft hydrogel semiconductors with augmented biointeractive functions

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Hydrogels, known for their mechanical and chemical similarity to biological tissues, are widely used in biotechnologies, whereas semiconductors provide advanced electronic and optoelectronic functionalities such as signal amplification, sensing, and photomodulation. Combining semiconducting properties with hydrogel designs can enhance biointeractive functions and intimacy at biointerfaces, but this is challenging owing to the low hydrophilicity of polymer semiconductors. We developed a solvent affinity–induced assembly method that incorporates water-insoluble polymer semiconductors into double-network hydrogels. These semiconductors exhibited tissue-level moduli as soft as 81 kilopascals, stretchability of 150% strain, and charge-carrier mobility up to 1.4 square centimeters per volt per second. When they are interfaced with biological tissues, their tissue-level modulus enables alleviated immune reactions. The hydrogel's high porosity enhances molecular interactions at semiconductor-biofluid interfaces, resulting in photomodulation with higher response and volumetric biosensing with higher sensitivity.

Hydrogels, featuring water-swollen three-dimensional (3D) polymer networks, have a soft, elastic, and water-abundant nature that emulates the mechanical and chemical environment of biological tissues (hereafter, biotissues), making them one of the optimal types of materials for interfacing with biotissues (*1*, *2*). With their great chemical tunability and controllable porous framework, hydrogels serve as a versatile platform for supporting, treating, and communicating with biotissues, enabling a broad range of biointerfaced applications including cell culture (*3*), tissue engineering (*4*), drug delivery (*5*), wound dressing (*6*), biostimulation (*7*), and biomonitoring (*8*).

An emerging technology is bioelectronics (*9*–*11*), which leverages the sophisticated and diverse functions of electronics to probe and program biology with high accuracy and resolution. The development of bioelectronics centers around the realization of seamless, dynamic, and lasting interfaces between tissue and electronics, for which hydrogels are an alternative to conventional electronic materials (e.g., conductors, semiconductors) for forming conformable tissue attachment (*12*), suppressing immune reactions (*13*), and facilitating mass transport (*14*). To fully exploit these capabilities of hydrogels, electronic and optoelectronic functions need to be embodied in hydrogel designs. Conductive hydrogels have been reported, and these are based on the design strategies of either swelled networks of conductive polymer (*15*) [mostly poly(3,4-ethylenedioxythiophene) polystyrene sulfonate, or PEDOT:PSS (*11*, *16*, *17*)] or hydrogel composites with conductive fillers such as metallic nanofillers (*18*) or 2D materials (*19*). However, the bioelectronic functions of such conductive hydrogels are limited to electrode-based stimulation and passive sensing with moderate sensitivity. In contrast, semiconductors, which feature electronic band structures with bandgaps, can enable much richer electronic and optoelectronic functions, including charge-transport modulation, amplification, photoexcitation, photo- and electroluminescence, and photocatalysis (Fig. 1A). Therefore, imparting hydrogels with semiconducting properties is a highly desirable but unmet need for enabling seamless biointerfaces to access a much broader range of advanced functions in sensing, stimulation, and imaging.

To realize semiconducting hydrogels with high electronic performance, polymer semiconductors (*20*, *21*) are arguably the most feasible choices for forming percolated 3D networks to achieve efficient charge transport. However, the main challenge comes from the incompatibility of the existing polymer semiconductors with the hydrogel preparation methods. Typically, hydrogels, including conductive hydrogels, are formed through chemically or physically crosslinking hydrogel precursors (i.e., either monomers or polymers) in water and remain in a highly water-swelled state. Such a process requires the aqueous processability of hydrogel precursors. However, for polymer semiconductors, the vast majority of existing designs are only soluble in organic solvents. Although certain physical processing (*22*, *23*) and chemical modification (*24*, *25*) approaches have been used to enable aqueous processability, they typically either sacrifice the electrical performance or limit the accessibility of different electronic structures for diverse optoelectronic functions. As a result, the reported hydrogel semiconductors exhibited charge-carrier mobility no higher than 0.25 $cm^2 V^{-1} s^{-1}$, with the functionality primarily limited to transistors (*25*).

We realized hydrogel semiconductors (hydro-SCs) by devising a processing method that decouples 3D network formation and water swelling into separate steps (Fig. 1B), which allows the processing of common types of polymer semiconductors to form gels in organic solvents. This processing method produces a morphology with a well-percolated network of polymer semiconductors embedded in a porous hydrogel framework.

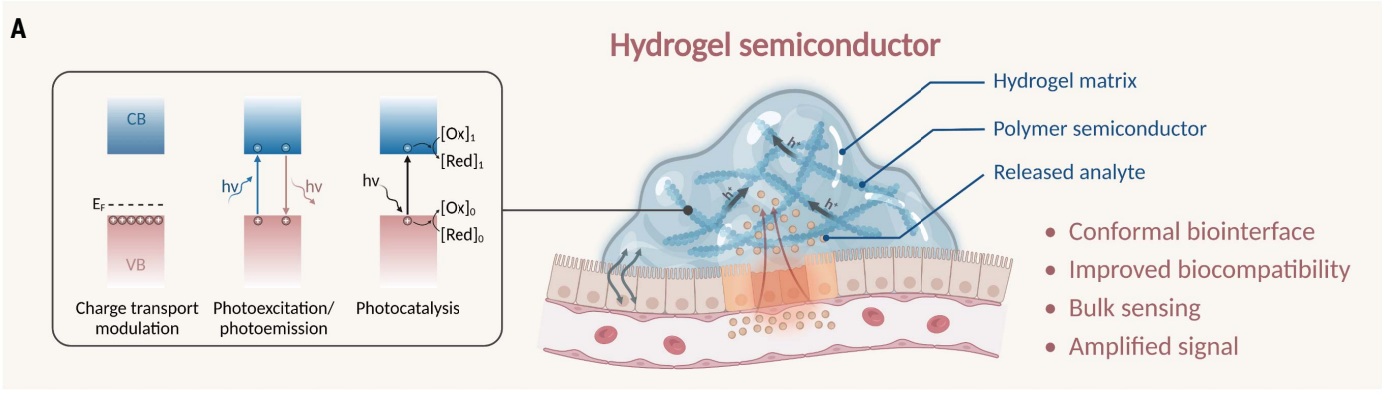
Design and preparation method for hydro-SCs

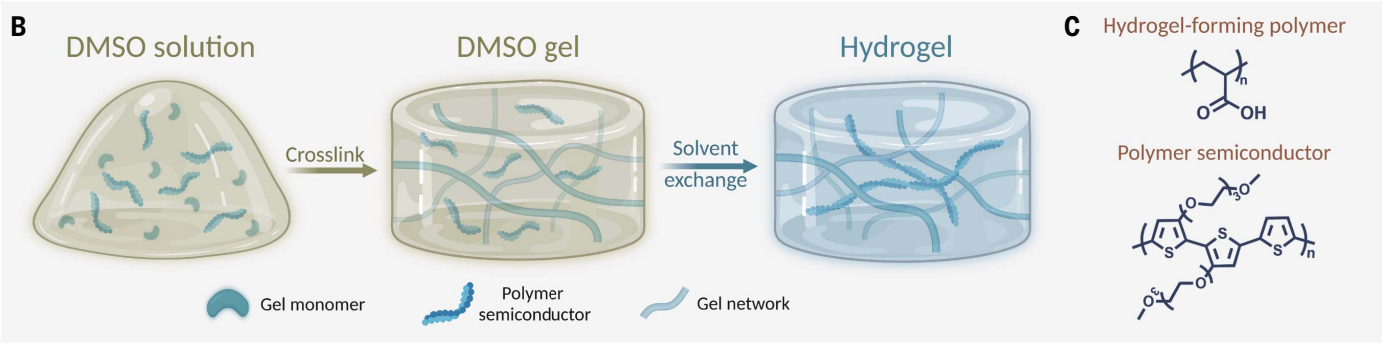
The design of a hydro-SC with an interpenetrated network morphology comprises two main components: a polymer semiconductor with hydrophilic side chains, and a hydrogel-forming polymer (Fig. 1B). Our method to generate a hydro-SC (Fig. 1B) starts with the preparation of a mixed solution of a polymer semiconductor and a hydrogel monomer in dimethyl sulfoxide (DMSO), together with the hydrogel cross-linker and photoinitiator. After spin coating or casting the mixed solution into the desired form factor, ultraviolet (UV) cross-linking is carried out to form the 3D network of the hydrogel polymer, thereby inducing gelation in DMSO. Finally, the gel is immersed in water to fully exchange DMSO with water, which turns the DMSO organogel into a hydrogel. During this process, as the polymer semiconductor is not soluble in water, we hypothesize that the semiconductor polymer will rapidly precipitate out and assemble into a percolated network inside the porous 3D hydrogel framework (Fig. 1B).

To implement this strategy, we chose poly(3,3′-bis{2-[2-(2-methoxyethoxy)ethoxy]ethoxy}2,2′:5′,2″-terthiophene), or p(g2T-T), as the polymer semiconductor (fig. S1) and acrylic acid (AAc) as the hydrogel-forming monomer for its high reactivity in DMSO (Fig. 1C). With initial concentrations of 20 and 200 mg/ml for p(g2T-T) and AAc, respectively, we successfully prepared hydro-SCs (figs. S2 to S5) that are very soft with Young's modulus as low as 81.8 kPa, exhibiting conformability on skin (Fig. 1D). When functioning as the channel layer in organic electrochemical transistors (OECTs) (Fig. 1E), the percolation of the semiconducting phase gives ideal OECT performance. Compared with soft and stretchable

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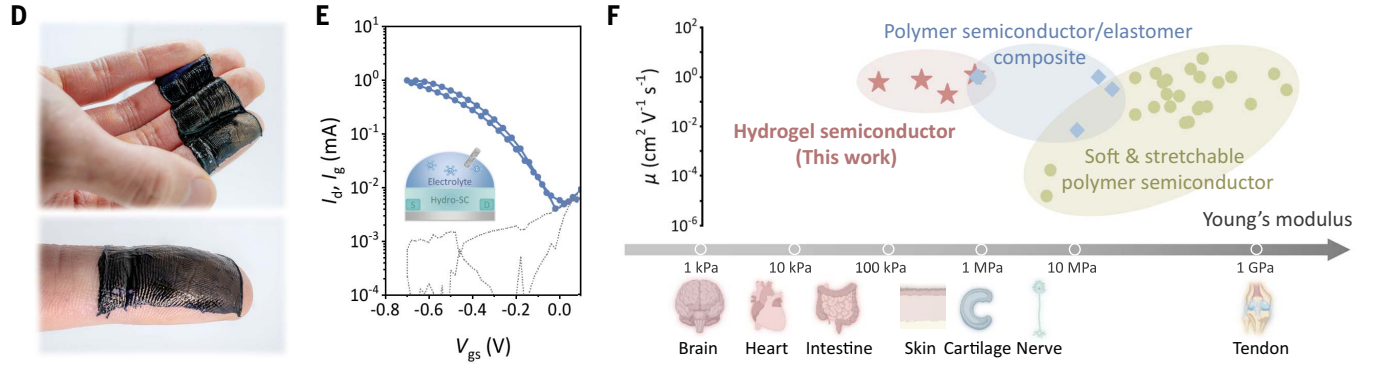


Fig. 1. Hydro-SC for biointerfaced multimodal applications. (**A**) Design of a hydro-SC based on an interpenetrated double network of a polymer semiconductor and a hydrogel-forming polymer. CB, conduction band; VB, valence band; *hν*, photon; Ox, oxidant; Red, reductant. (**B**) Preparation method of the hydro-SC, which decouples the gelation step and water swelling step through the solvent exchange. (**C**) Chemical structures of the hydrogel-forming polymer and polymer semiconductor used: PAAc and p(g2T-T), respectively. (**D**) Photograph showing a prepared hydro-SC conforming on the skin. (**E**) Hydro-SC film serving as the channel layer in an OECT, giving normal transfer behavior (V_{ds} = −0.8 V). The blue and gray curves refer to I_d and I_g, respectively. (**F**) Comparisons of Young's modulus and charge-carrier mobility from the hydro-SCs (dark red) with other reported soft and stretchable semiconductors (with the reference information shown in fig. S6 and table S1).

polymer semiconductors, this hydro-SC realizes a modulus decrease of more than two to three orders of magnitude while keeping the mobility (μ) at a similar level (Fig. 1F, fig. S6, and table S1). Moreover, such a strategy can be applied to a number of polymer semiconductors with polar side chains (fig. S7) and can enable direct photopatterning for high-density device fabrication (fig. S8).

Characterization of the morphological evolution during hydro-SC preparation

The key part of the design and preparation of the hydro-SC is the assembly process of the polymer semiconductor p(g2T-T) during the gelation and solvent exchange steps. To trace the morphological evolution, we characterized three stages during the preparation process: the initial DMSO-based solution as stage 1, the UV cross-linked gel in DMSO as stage 2, and the final hydro-SC after the solvent exchange as stage 3 (Fig. 2A). The hydro-SC was prepared with a weight ratio of 10/100 between p(g2T-T) and AAc, and the added cross-linker of 0.2 wt % to the AAc monomer. For stage 1, cryo–electron microscopy (cryo-EM) imaging of the rapidly frozen DMSO-based solution shows that p(g2T-T) already formed high-density nano-aggregates (Fig. 2B, stage 1), likely because of the limited solvation effect of DMSO only on the polar side chains and not the hydrophobic backbones of p(g2T-T) (*26*) (fig. S9). These preformed nanoaggregates could help to seed the further assembly of p(g2T-T) into larger networks. For stage 2, scanning electron microscopy (SEM) and scanning transmission electron microscopy (STEM) imaging on a DMSO gel reveals that p(g2T-T) particles mostly maintain their size (Fig. 2B, stage 2, and fig. S10). Finally, water solvent exchange induces the assembly of p(g2T-T) into an interconnected network that is interpenetrated

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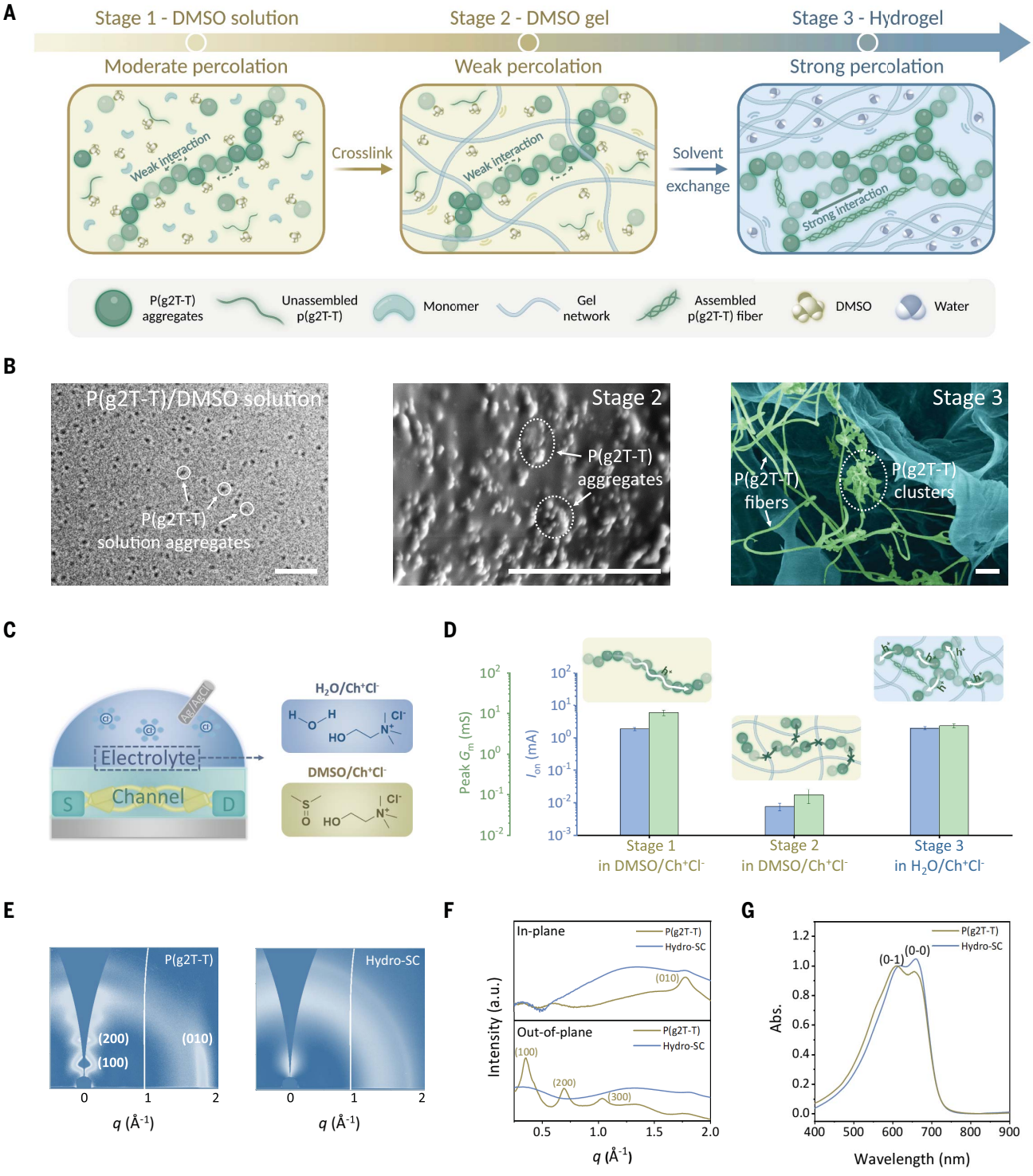


Fig. 2. Morphological evolution during the formation of a hydro-SC. (**A**) Schematics showing the morphology evolvement during the three stages of the hydro-SC preparation, highlighting the aggregation and network-formation process of p(g2T-T). (**B**) Cryo-EM (stage 1 but without AAc monomers) and SEM images (stages 2 and 3) of the preparation process for hydro-SC at different stages (scale bars, 2 μm), showing the existence of p(g2T-T) aggregates in the initial DMSO solution and DMSO gel, and the formation of the interconnected network after solvent exchange in water. The image of stage 3 was colored with green and blue representing the p(g2T-T) and PAAc network, which were identified on the basis of the mass ratio and porosity. (**C** and **D**) OECT tests in two types of electrolytes (i.e., Ch^+Cl^- in DMSO, and Ch^+Cl^- in water) (C), and electrical performance of hydro-SC at each of the preparation stages (D), revealing the molecular-level percolation of p(g2T-T) (n = 3 independent measurements). (**E**) 2D GIXD images of a neat p(g2T-T) film and a hydro-SC film. (**F**) 1D linecuts in the in-plane and out-of-plane directions of the neat p(g2T-T) and hydro-SC film. a.u., arbitrary units. (**G**) Normalized UV-vis spectra of a neat p(g2T-T) film and a hydro-SC film. Both films were electrochemically dedoped to −0.2 V versus Ag/AgCl during the measurement. Abs., absorbance.

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with the porous hydrogel (i.e., PAAc) framework. Such a morphology is shown by SEM (Fig. 2B, stage 3, and fig. S11) on a freeze-dried sample and also validated by other microscopic imaging in different length scales, including dark-field microscopy (fig. S12), conductive atomic force microscopy (C-AFM) (fig. S13), and STEM (fig. S14) on the wet or dried gel samples. This morphological evolution process validated our hypothesis that the exchange to a nonsolvent (i.e., water) drives the assembly of polymer semiconductor chains into a percolated network for enabling long-range charge transport.

We further measured the charge-transport performance by using each stage of the solution or gel to make OECT devices. By performing our hydrogel preparation process on substrates with predeposited gold electrodes, the semiconductor solution and gel at each stage can serve as the OECT channel (Fig. 2C). For stages 1 and 2, we use an electrolyte with DMSO solvent and organic choline chloride (Ch^+Cl^-) salt, which was added on top of the DMSO solution and gel in these two stages to enable sufficient ion penetration into the semiconducting channel layers (fig. S15). Stage 1 already gave high electrical performance with a transconductance (G_m) of 8.54 mS (Fig. 2D and fig. S16A), indicating that molecular-level percolation of p(g2T-T) occurs in the solution state in DMSO. After the UV cross-linking of AAc monomers in DMSO (stage 2), the electrical performance decreased by about two orders of magnitude (Fig. 2D and fig. S16B), suggesting that the formation of the PAAc 3D network breaks the molecular-level percolation of p(g2T-T). After the solvent exchange, the formation of the microscopic p(g2T-T) fiber network in the final hydro-SC reverted the electrical performance to a similar level with stage 1, as shown by the OECT measurement (Fig. 2D and fig. S16C) using an aqueous choline chloride electrolyte. Such a performance evolution also aligns with the transition of short-range ordering during the hydro-SC preparation, which is evidenced by UV-vis (visible) spectra (fig. S17).

For the hydro-SC thin film, we further studied the polymer packing of p(g2T-T). Grazing-incidence x-ray diffraction (GIXD) (Fig. 2E) shows that, in contrast to a neat p(g2T-T) film deposited from a chloroform solution, the hydro-SC film does not have observable crystalline features (Fig. 2, E and F). This suggests that the formation of long-range ordering is suppressed during the rapid polymer chain precipitation from the solvent exchange process, together with the inhibition effect of the PAAc network. On the other hand, UV-vis absorption spectroscopy reveals enhanced short-range aggregation in hydro-SC films (Fig. 2G), as evidenced by the higher (0-0)/(0-1) ratio than that from p(g2T-T) films. We reason that the higher aggregation level comes from the stronger aggregation of p(g2T-T) in DMSO compared with the dissolution in chloroform. Overall, such a packing morphology could be an optimal balance for the hydro-SC with dual design goals: On the one hand, the enhanced local aggregation and the microstructure-level percolation can ensure efficient interchain charge transport, and on the other hand, the lack of crystallization can help to achieve softer mechanical properties. For the electronic structures, we also used cyclic voltammetry (fig. S18) and UV-vis absorption to confirm that the HOMO (highest occupied molecular orbital) and LUMO (lowest unoccupied molecular orbital) levels of the hydro-SC are almost identical to those of the neat p(g2T-T), which give similar redox activities under electrochemical biasing (fig. S19).

Electrical and mechanical properties of hydro-SCs with varied compositions

For the hydro-SC design, there are two main compositional factors that can influence both electrical and mechanical properties: (i) the cross-linking density of the PAAc network and (ii) the weight ratio between p(g2T-T) and PAAc. We first compared the electrical performance in OECT devices (Fig. 3A). The influence of cross-linking density was tested by varying the cross-linker weight percentage among 0.2, 0.5, and 2% of the AAc monomer, with the p(g2T-T)-to-AAc weight ratio kept at 10/100. With all three compositions showing ideal OECT transfer and output behaviors (Fig. 3B and fig. S20), the lowest cross-linking ratio gives the highest μ of 1.22 $cm^2 V^{-1} s^{-1}$ (Fig. 3C and fig. S21), which could be attributed to the more ordered assembly of p(g2T-T) in a looser PAAc network. The slightly higher μ from 2% cross-linking than from 0.5% could come from the lower swellability of the hydrogel with a higher cross-linking ratio (fig. S22). This is evidenced by the comparison of the OECT performance from the hydro-SC in the dry state, using ion-exchange gel as the electrolyte (*27*) (fig. S23). On the other hand, when we increase the p(g2T-T)-to-PAAc weight ratio from 5/100 to 13.3/100 while the cross-linker concentration is kept at 0.2% (fig. S24 and table S2), we see a substantial increase of the μ from a nonmeasurable level to 1.43 $cm^2 V^{-1} s^{-1}$ (Fig. 3D and fig. S25), which should come from a better percolation of the p(g2T-T) network. Further increase of the p(g2T-T) ratio gives higher μ toward the neat p(g2T-T) (fig. S26). Overall, these trends in μ are consistent with the changes in aggregation levels with the varied compositional factors, as revealed by the UV-vis absorption spectra (fig. S27). The volumetric capacitance (C^*) of the p(g2T-T) within the hydro-SC is similar to that of the pristine p(g2T-T) film (fig. S28).

We studied the mechanical properties of the hydro-SC thin film (table S3) through a floating-on-water (FOW) tensile test (*28*) (Fig. 3E and movie S1). We used two types of supporting liquid, a 100 mM NaCl aqueous solution and phosphate-buffered saline (PBS), for the films to reach the same swelling states with the OECT tests and the physiological environment, respectively. Such an electrolyte environment will lead to less swelling and slightly higher modulus compared with water. As shown in Fig. 3F and figs. S29 and S30, the gradual increase of the cross-linking density first leads to a decrease and then an increase of the modulus, which should come from the combined effect of a denser network and higher swellability. The increased weight ratio of p(g2T-T) consistently gives a high modulus simply because of the increased fraction of the more rigid component (Fig. 3G and fig. S31). For the lowest modulus of 81 kPa achieved from hydro-SCs, it is a decrease by more than three orders of magnitude than the neat p(g2T-T) (Fig. 3G). The hydro-SC also exhibits much higher stretchability than p(g2T-T) (fig. S32). Supported on polystyrene-*block*-poly(ethylene-*ran*-butylene)-*block*-polystyrene (SEBS) substrates, hydro-SC (0.5% cross-linking, 10/100 ratio) films can be stretched up to 150% strain without crack formation or OECT performance degradation (Fig. 3H and fig. S33). Such mechanical and electrical stability also extends to repeated bending (fig. S34) and physiological conditions (fig. S35).

Hydro-SCs for biointerfacing: Higher conformability and lower immune reactions

The first advantage of using hydro-SCs to interface with biotissues or skin is the higher conformability from the ultralow modulus (*29*). As demonstrated by adhering a 10-μm-thick hydro-SC film (10/100 and 0.5% cross-linking hydro-SC was used unless specified) to a tendon replica with a rough and irregular surface (fig. S36A), the film adapted to the surface morphology (figs. S36B and S37A), showing a clear improvement compared with a neat p(g2T-T) film, which is much thinner (~650-nm-thick) (figs. S36C and S37B).

The second advantage of the tissue-level modulus of hydro-SC thin films is the alleviated foreign body response (FBR) (*30*). This was tested by coating different films on both sides of SEBS substrates and implanting subcutaneously in mice for 4 weeks. From the explanted tissues around the implants (figs. S38 and S39), Masson's trichrome staining (Fig. 4A and fig. S38) shows that the hydro-SC films with 2.0% cross-linker and a modulus of 239 kPa had reduced collagen densities (Fig. 4B) compared with both the bare SEBS implants and the p(g2T-T) films. The level of collagen density is already comparable to that of PAAc hydrogel, which has a modulus of 12.8 kPa (fig. S40A). Such an effect of Young's modulus on the FBR was further validated by hydro-SC films

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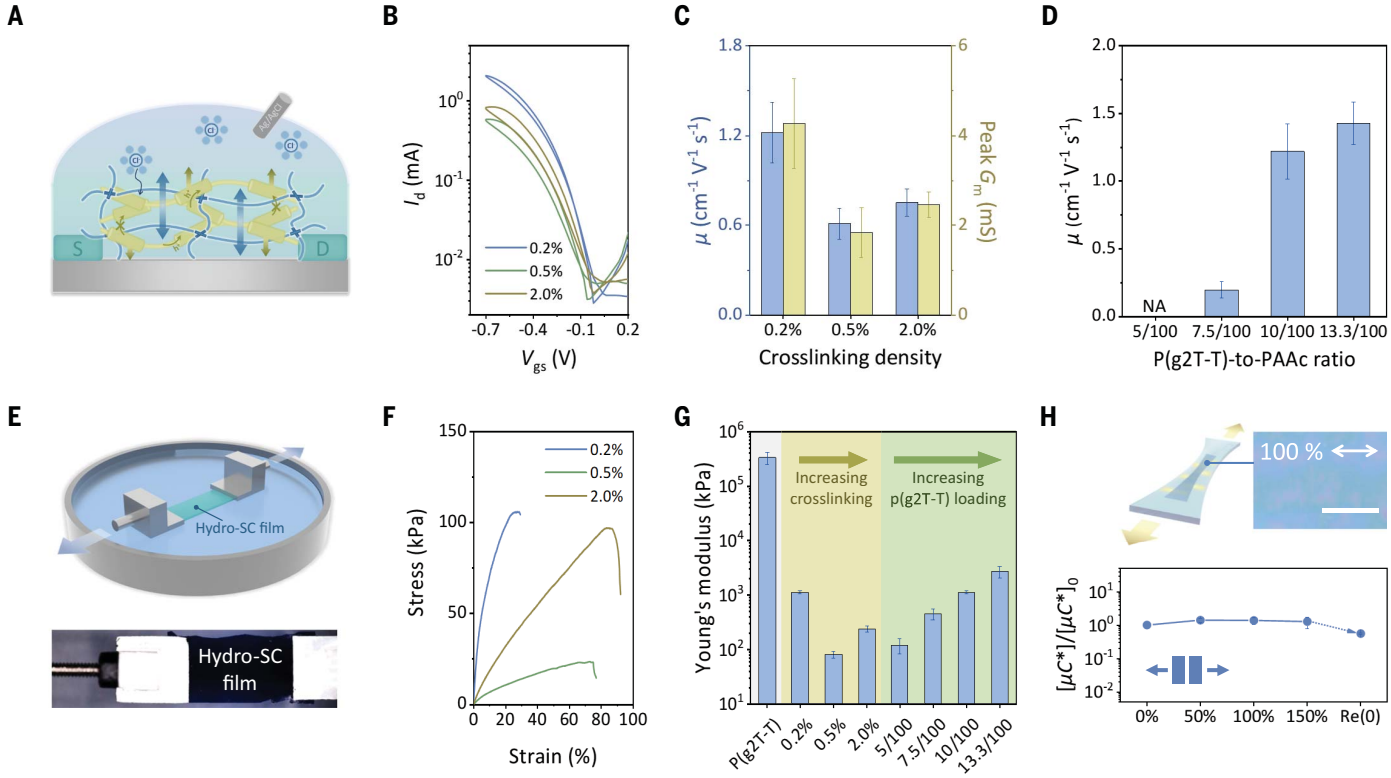


Fig. 3. Electrical and mechanical properties of hydro-SC films. (**A**) OECT tests using swelled hydro-SC films. (**B** and **C**) OECT transfer curves (B), extracted μ, and peak G_m (C) from hydro-SCs with three different cross-linking ratios (V_{ds} = −0.8 V) (n = 9 independent measurements). (**D**) Extracted μ from hydro-SCs with different p(g2T-T)-to-AAc weight ratios (n = 9 independent measurements). (**E**) Schematic (top) and picture (bottom) showing the FOW test for the mechanical properties of hydro-SC films. (**F**) Stress-strain curves of hydro-SC films with different cross-linking ratios measured in 100 mM NaCl solution. (**G**) Young's modulus of p(g2T-T) and hydro-SC films with different p(g2T-T)-to-AAc weight ratios (n = 4 independent measurements). (**H**) Stretchability test of hydro-SC films on a SEBS substrate with source-drain electrodes (top), with the optical microscopy image showing the morphology under 100% strain (scale bar, 20 μm), and normalized μC^* from OECT measurements during the stretching to 150% strain (in parallel to the charge-transport direction) and release (n = 9 independent measurements).

with much higher cross-linking ratios (10%) and thereby a higher modulus of 4.76 MPa (fig. S40B). The collagen density level is higher than in the softer hydro-SC and actually stays on a similar level with the neat p(g2T-T) films (Fig. 4B). This trend is also observed from the comparison of myofibroblast deposition shown by immunofluorescence imaging (fig. S41). Such an influencing behavior of surface modulus on the FBR level is consistent with previous studies on hydrogels and elastomers (*31*, *32*).

Hydro-SCs display enhanced photomodulation effects

Semiconductors have been demonstrated with photomodulation functions for biointerface applications, through photoelectrochemical or photothermal effects. We first studied the photoelectrochemical behavior (Fig. 4C) of our hydro-SC using an electrochemical cell setup under physiological conditions (Fig. 4D). When irradiated by a 625-nm laser, the hydro-SC film showed a lasting and stable faradaic current, which indicates the charge transfer happening between the semiconductor and the electrolyte (Fig. 4E and fig. S42). The stabilized current level is substantially higher than that of a neat p(g2T-T) film [of the same weight as the p(g2T-T) component in the hydro-SC film] tested under the same conditions (Fig. 4E). We reason that this is mostly attributable to the porous nature of the hydrogel design, which substantially increases the semiconductor-electrolyte interfaces for more-efficient use of electrons and holes by redox reactions (*33*).

We next studied the photothermal properties that come from the nonradiative decay after photoexcitation (Fig. 4F). Upon a laser radiation power of 290 mW cm^{-2}, a hydro-SC film reached 78°C within 30 s (fig. S43A). Under the same radiation condition of 178 mW cm^{-2}, the hydro-SC film achieved more-efficient heat accumulation (Fig. 4G and fig. S43B), giving a higher photothermal conversion efficiency (PCE) (see supplementary text in the supplementary materials) than a neat p(g2T-T) film (Fig. 4H, figs. S44 and S45, and table S4). When applied on biotissue (e.g., pig skin) surfaces covered with biofluids, the hydro-SC in the fully swelled state can conformally adhere to the tissue and also generate rapid heating with the temperature reaching 50°C in 60 s (Fig. 4I and fig. S46), under a radiation power of 318 mW cm^{-2} (movie S2). As such, for therapeutic applications, our hydro-SC could serve as a dressing layer for wound healing processes, combining efficient photothermal treatment, soft modulus, oxygen permeability, and moisturization of wound surfaces (*6*).

Bulk diffusion–enabled volumetric biosensing

Another key biointerfaced application for the hydro-SC is biochemical sensing, for example, through OECT devices. Biochemical sensing can particularly benefit from the bulk mass transport properties of hydrogels. For existing polymer semiconductors, the dense nature and limited hydrophilicity can hardly allow the transport of biomolecules into the bulk volume. As a result, both the functionalization of bioreceptors and the sensing of bioanalytes can only happen at the surfaces (Fig. 5A). This has been the sensing mode for all the transistor-type biosensors (*34*–*37*). For our hydro-SC, the mass transport properties for biomolecules now can open up the possibility for volumetric sensing (Fig. 5A) throughout the entire thickness of the semiconducting layer. We reason

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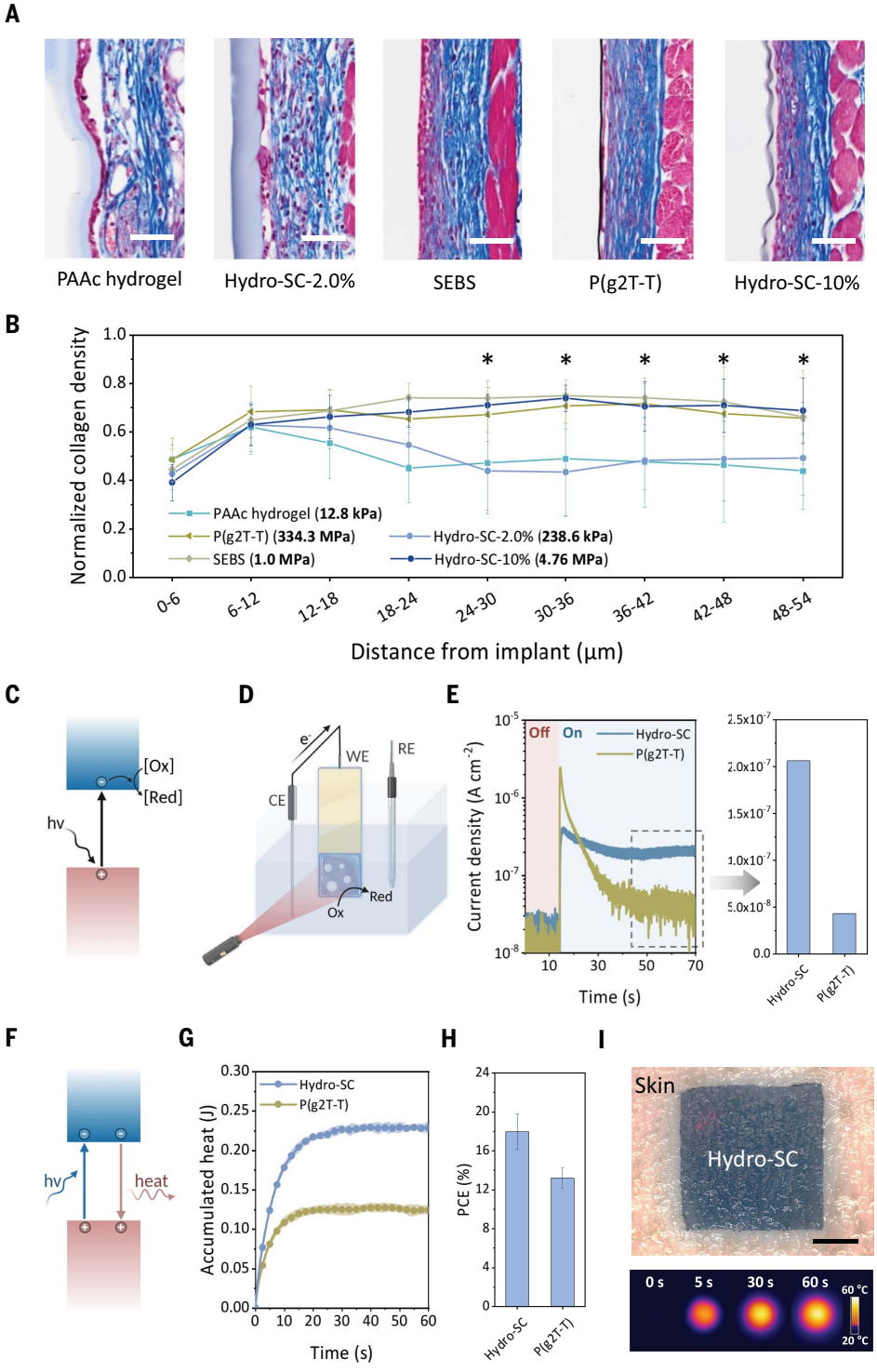


Fig. 4. Hydro-SCs for alleviated FBR and enhanced photomodulation effects. (**A**) Masson's trichrome staining results of tissues around five types of implants after 4 weeks of subcutaneous implantation in mice. Scale bars, 50 μm. (**B**) Normalized collagen density of different implants as a function of distance (n = 6 independent measurements). Statistical significance and P values were determined by two-sided Student's t test; P values were calculated between hydro-SC 2.0% and p(g2T-T) samples. *P < 0.05. (**C**) Mechanism of the photoelectrochemical effect from semiconductors. (**D**) Schematic showing the test setup for photoelectrochemical measurement. CE, counter electrode; WE, working electrode; RE, reference electrode. (**E**) Photoelectrochemical response of hydro-SC and p(g2T-T) films under a radiation power of 114.3 mW cm^{-2} (left), and comparison of photofaradaic current density (right). The amount of p(g2T-T) was kept the same in both films. Tests were conducted with a 0 V bias applied versus the Ag/AgCl electrode. The positive photocurrent values stand for the electrons flowing out of the tested films, and the baseline current was subtracted. (**F**) Mechanism of the photothermal effect from semiconductors. (**G**) Accumulated heat on the hydro-SC or neat p(g2T-T) films and the surrounding substrates, upon a radiation power of 178 mW cm^{-2}. The amount of p(g2T-T) was kept the same in both films. (**H**) Calculated PCE of the two films (n = 3 independent measurements). (**I**) Photograph showing a hydro-SC film conformably adhered to a piece of wet pig skin (top), and thermal heating from the hydro-SC film applied on the skin (bottom). Radiation power, 318 mW cm^{-2}. Scale bar, 5 mm.

that such volumetric sensing can enable many more interaction sites between bioreceptors and analyte molecules and thereby higher sensitivity. To test this possibility, we first used fluorescent imaging to directly visualize the diffusion ability of large-sized molecules through a hydro-SC film–based membrane (Fig. 5B). This experiment was carried out using a macromolecular fluorescent probe PEG-FITC (polyethylene glycol–fluorescein isothiocyanate) [number-average molecular weight (M_n) = 20 kDa; fig. S47], which emulates the molecular sizes of proteins and nucleic acids. As shown in Fig. 5C and fig. S48, the fluorescent probes, initially on one side of the hydro-SC membrane (~5-μm-thick), diffused to the other side after 24 hours. In comparison, for a membrane made from a neat p(g2T-T) film, although much thinner (~500-nm-thick), such diffusion was not observed (Fig. 5C and fig. S49).

To further realize volumetric biosensing, a key requirement is to have a method to functionalize bioreceptors throughout the bulk of hydro-SC. We designed an approach that

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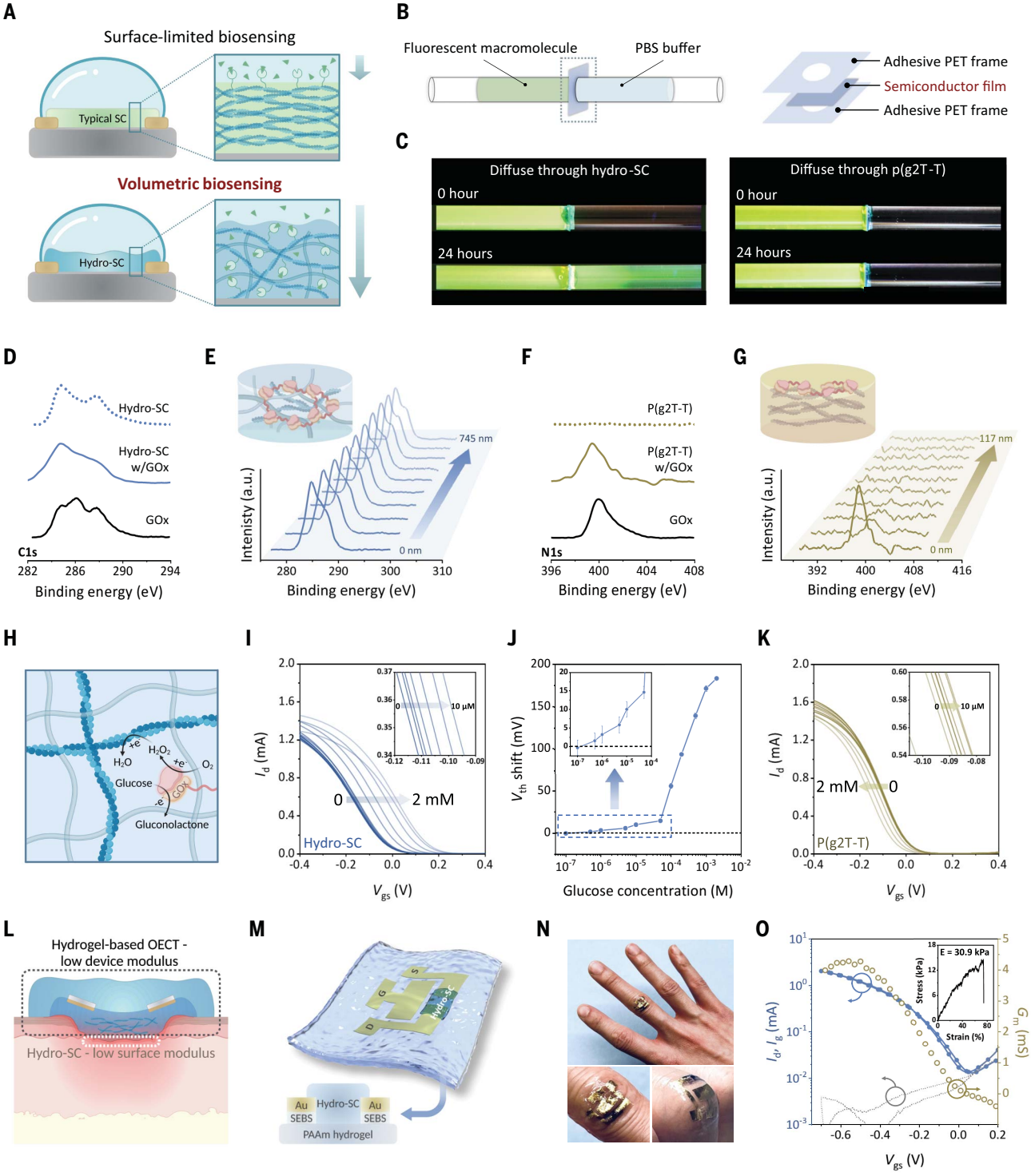


Fig. 5. Biosensing using hydro-SC films: Volumetric sensing with higher sensitivity and hydrogel-based OECT devices with tissue-like moduli. (**A**) Schematics comparing the mechanism of conventional surface-limited biosensing (top) and volumetric biosensing (bottom), showing that volumetric biosensing provides more interaction sites with analytes. (**B**) Setup for qualitatively testing the diffusion ability of semiconductor films by large molecules, in which a semiconductor film serves as the membrane with fluorescent probe molecules initially only on one side of the membrane. (**C**) Diffusion of fluorescent probes through hydro-SC (left) and neat p(g2T-T) (right) films in 24 hours. (**D**) C1s spectra of XPS characterizations on GOx alone, a hydro-SC film without GOx immobilization, and a hydro-SC film with GOx immobilization. (**E**) C1s spectra of XPS depth profiling through the thickness of a hydro-SC film after GOx immobilization. (**F**) N1s spectra of XPS characterizations on GOx alone, a neat p(g2T-T) film without GOx immobilization, and a neat p(g2T-T) film with GOx immobilization. (**G**) N1s spectra of XPS depth profiling through the thickness of a neat p(g2T-T) film after GOx immobilization. The GOx was immobilized with a surface coverage of 58% (fig. S57), through noncovalent interactions between GOx and PEG side chains. (**H**) Sensing response mechanism in hydro-SC during volumetric sensing of glucose. (**I**) Volumetric sensing performance for glucose from a hydro-SC as the channel layer in an

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OECT device. (**J**) Correlation between the threshold voltage (V_{th}) shift and glucose concentration (n = 3 independent measurements). (**K**) Surface sensing performance for glucose from a neat p(g2T-T) film as the channel layer in an OECT device. (**L**) Schematic showing a hydrogel-based OECT (hydro-OECT) for interfacing with biotissues with tissue-like moduli both on the device surface and from the whole device. (**M**) Device structure of a hydro-OECT. (**N**) Photographs of a hydro-OECT device on a bending finger joint, showing highly conformable contact. (**O**) Modulus measurement (inset) and electrical performance of a hydro-OECT. V_{ds} = −0.8 V. The transfer curve was acquired with a scan speed of 10 mV s^{-1}.

uses cross-linking to immobilize the penetrated bioreceptor molecules in the hydro-SC layer (fig. S50). Specifically, one possible chemistry to realize such cross-linking is by using glutaraldehyde as the cross-linker to react with amines that universally exist in proteins (*38*). To demonstrate this approach, we used glucose oxidase (GOx) as an enzymatic bioreceptor (fig. S51). The hydro-SC used in this experiment was prepared with poly(acrylamide) (PAAm) as the hydrogel-forming polymer because of its electrical neutrality (fig. S52). After 24 hours of incubation in GOx solution with the cross-linker, the successful immobilization of GOx throughout the thickness of the hydro-SC film could be confirmed through x-ray photoelectron spectroscopy (XPS) depth profiling (Fig. 5, D and E) on samples after washing and drying. In this analysis, the C1s peak from GOx was used instead of the stronger N1s peak, as PAAm also has the nitrogen element (Fig. 5D and fig. S53). The GOx density is determined as 782 nmol cm^{-3} by the quartz-crystal microbalance (QCM) (table S5). In contrast, the same functionalization process performed on a neat p(g2T-T) only leads to the immobilization of GOx on the surface (Fig. 5, F and G), which is due to the limited infiltration of GOx into the p(g2T-T) film.

We further tested the volumetric sensing performance of the bulk-functionalized hydro-SC in OECT devices, which was compared with surface-functionalized p(g2T-T) films. For glucose analyte, the sensing can happen through glucose oxidation at the active site of GOx, which leads to the reduction of oxygen to hydrogen peroxide that further oxidizes the p(g2T-T) (Fig. 5H). For hydro-SC, the gradual increase of the drain current (I_d) can be observed during the increase of the glucose concentration from 100 nM to 2 mM (Fig. 5I). By analyzing the relationship between the shift of threshold voltage (V_{th}) with the glucose concentration (Fig. 5J), the detection limit can be determined as 500 nM. In comparison, the surface-functionalized p(g2T-T) only shows a current decrease when the glucose concentration is >100 μM (Fig. 5K), which could come from the electrooxidation of glucose (*39*) (fig. S54). This indicates that surface-immobilized GOx might be insufficient to achieve considerable enzymatic reactions (fig. S55). Overall, this comparison validated that the volumetric biosensing achieved on the hydro-SC can be an effective strategy for increasing sensitivity. The glucose sensor also generated a response in more-realistic biological fluids such as serum and sweat (fig. S56), showing promise for in vivo sensing applications.

Hydrogel-based OECTs

For the application of such OECT-based biosensing with direct interfacing with biotissues, the requirement on tissue-level softness is mainly composed of two aspects: the surface modulus that is in direct contact with the tissue, and the effective modulus of the full device (Fig. 5L). The hydro-SC as the sensing surface can satisfy the former requirement. To fulfill the latter part, we used the hydro-SC to create stretchable OECT devices on a hydrogel substrate (Fig. 5M and fig. S58). The stretchable electrodes were served by microcracked gold films (*40*), which were deposited on a thin layer of SEBS sitting on top of the hydrogel substrate. The SEBS layer not only forms an interpenetrated interface with the gold films to enable the stretching-induced microcrack morphology but also prevents shorting of the electrodes by water in the hydrogel. The obtained hydrogel-based OECT gives high conformability even on a highly wrinkled part of the skin (Fig. 5N), which benefits from the ultralow modulus of ~30.9 kPa (Fig. 5O, inset). The OECT shows transistor performance (Fig. 5O), with a high on-current of 2 mA, on-off ratio of 10^2, and decent operation speed (fig. S59). In contrast, when a neat p(g2T-T) film was used as the channel layer to make a hydrogel-based OECT, the substantial modulus mismatch with the hydrogel substrate led to very low mechanical robustness (fig. S60).

Our interpenetrated double-network design for hydrogel semiconductors unites high-performance semiconducting properties with tissue-like softness, high porosity, and efficient transport of biomolecules. The hydrogel properties improve the conformability and immune compatibility for the use of semiconductors on tissue surfaces. The highly porous morphology, with a high surface aspect ratio, also enhances the bioelectronic functions, such as stronger photoelectrochemical and photothermal effects for photomodulation, and biosensing with higher sensitivity through volumetric sensing. All these are enabled by our solvent-exchange processing method, which circumvents the need for water processability for polymer semiconductors. This method offers broad applicability to different designs of redox-active semiconducting polymers, so that hydrogel designs can be achieved on different semiconductor-based functions that need different band structures and optoelectronic properties. We envision that hydrogel semiconductors that blur the mechanical and chemical interfaces between semiconductors and biotissues can play a central role in bridging electronics technologies with hydrogel-based biomedical technologies.

REFERENCES AND NOTES

1. H. Yuk, B. Lu, X. Zhao, *Chem. Soc. Rev.* **48**, 1642–1667 (2019).
2. A. C. Daly, L. Riley, T. Segura, J. A. Burdick, *Nat. Rev. Mater.* **5**, 20–43 (2020).
3. J. Lou, D. J. Mooney, *Nat. Rev. Chem.* **6**, 726–744 (2022).
4. K. Y. Lee, D. J. Mooney, *Chem. Rev.* **101**, 1869–1879 (2001).
5. J. Shi *et al*., *Nat. Chem. Eng.* **1**, 73–86 (2024).
6. Y. Liang, J. He, B. Guo, *ACS Nano* **15**, 12687–12722 (2021).
7. J. Deng *et al*., *Nat. Mater.* **20**, 229–236 (2021).
8. J. Ma *et al*., *Chem. Eng. J.* **481**, 148317 (2024).
9. P. Li *et al*., *Nature* **626**, 990–998 (2024).
10. J. Liu *et al*., *Science* **383**, 1096–1103 (2024).
11. X. Strakosas *et al*., *Science* **379**, 795–802 (2023).
12. H. Yuk, J. Wu, X. Zhao, *Nat. Rev. Mater.* **7**, 935–952 (2022).
13. S. Naahidi *et al*., *Biotechnol. Adv.* **35**, 530–544 (2017).
14. E. Axpe *et al*., *Macromolecules* **52**, 6889–6897 (2019).
15. D. Liu *et al*., *Mater. Horiz.* **10**, 2800–2823 (2023).
16. B. Lu *et al*., *Nat. Commun.* **10**, 1043 (2019).
17. T. Zhou *et al*., *Nat. Mater.* **22**, 895–902 (2023).
18. T. Dvir *et al*., *Nat. Nanotechnol.* **6**, 720–725 (2011).
19. Y. Lu *et al*., *Nat. Electron.* **7**, 51–65 (2024).
20. Y. Zheng, S. Zhang, J. B.-H. Tok, Z. Bao, *J. Am. Chem. Soc.* **144**, 4699–4715 (2022).
21. L. Ding *et al*., *Chem. Rev.* **123**, 7421–7497 (2023).
22. K. M. Sim *et al*., *ACS Nano* **14**, 415–421 (2020).
23. J. Cho *et al*., *Energy Environ. Sci.* **10**, 2324–2333 (2017).
24. B. Kang, Z. Wu, M. J. Kim, H. Y. Woo, J. H. Cho, *Chem. Mater.* **32**, 1111–1119 (2020).
25. P. Li *et al*., *Science* **384**, 557–563 (2024).
26. Z. Xu *et al*., *Adv. Mater.* **34**, e2203055 (2022).
27. C. G. Bischak, L. Q. Flagg, D. S. Ginger, *Adv. Mater.* **32**, e2002610 (2020).
28. L. A. Galuska *et al*., *Nat. Commun.* **12**, 2347 (2021).
29. Z. Yan *et al*., *Science* **375**, 852–859 (2022).
30. S. Capuani, G. Malgir, C. Y. X. Chua, A. Grattoni, *Bioeng. Transl. Med.* **7**, e10300 (2022).
31. N. Li *et al*., *Science* **381**, 686–693 (2023).
32. R. Sridharan, B. Cavanagh, A. R. Cameron, D. J. Kelly, F. J. O'Brien, *Acta Biomater.* **89**, 47–59 (2019).
33. C.-L. Chang *et al*., *Nat. Commun.* **13**, 5460 (2022).
34. N. Wang, A. Yang, Y. Fu, Y. Li, F. Yan, *Acc. Chem. Res.* **52**, 277–287 (2019).
35. X. Ji, X. Lin, J. Rivnay, *Nat. Commun.* **14**, 1665 (2023).
36. G. E. Fenoy *et al*., *ACS Appl. Mater. Interfaces* **15**, 10885–10896 (2023).
37. D. Ohayon *et al*., *Nat. Mater.* **19**, 456–463 (2020).
38. I. Migneault, C. Dartiguenave, M. J. Bertrand, K. C. Waldron, *Biotechniques* **37**, 790–802 (2004).
39. M. Pasta, F. La Mantia, Y. Cui, *Electrochim. Acta* **55**, 5561–5568 (2010).
40. Y. Jiang *et al*., *Nature* **614**, 456–462 (2023).

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SUPPLEMENTARY MATERIALS

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Soft hydrogel semiconductors with augmented biointeractive functions

Yahao Dai, Shinya Wai, Pengju Li, Naisong Shan, Zhiqiang Cao, Yang Li, Yunfei Wang, Youdi Liu, Wei Liu, Kan Tang, Yuzi Liu, Muchuan Hua, Songsong Li, Nan Li, Shivani Chatterji, H. Christopher Fry, Sean Lee, Cheng Zhang, Max Weires, Sean Sutyak, Jiuyun Shi, Chenhui Zhu, Jie Xu, Xiaodan Gu, Bozhi Tian, and Sihong Wang

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Editor's summary

Water-laden hydrogels are increasingly finding use in biomedical applications because of their inherent softness, which matches that of biological tissue, and their ability to maintain hydration and oxygen permeation. However, it is difficult to give them semiconducting functionality, as most polymer semiconductors are not soluble in water. Dai *et al*. report the design, synthesis, characterization, and applications of hydrogel–semiconducting polymer composites through a process of solvent exchange. They combined the polymer semiconductor p(g2T-T) with the hydrogel-forming monomer acrylic acid in dimethyl sulfoxide to form a double-network structure upon cross-linking with ultraviolet light. After solvent exchange with water, the semiconducting hydrogel had a low modulus while maintaining high charge carrier transport properties, as well as organ conformability and reduced foreign body response. —Marc S. Lavine

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